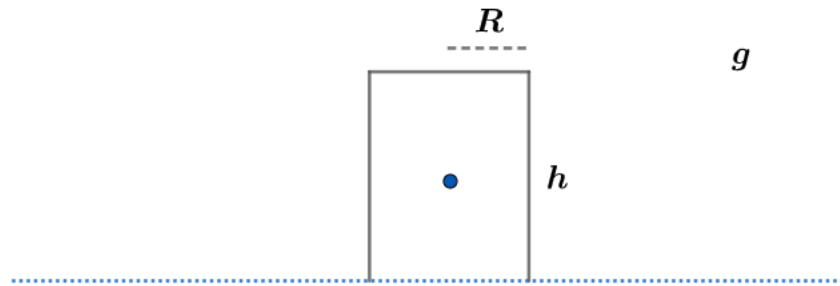


2021 F=ma Exam: Problem 6

Kevin S. Huang



By energy conservation, we increased the gravitational potential energy of the water by $mg(h/2)$ (center of mass raised by $h/2$) which is the work needed,

$$W = \frac{mgh}{2} = \frac{(\rho\pi R^2 h)gh}{2} = \frac{\rho g \pi R^2 h^2}{2}$$

so the answer is D.